KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: ELECTRONICS

DATE: 23rd February 2022 **TIME: 3 HOURS (9:00AM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

ELECTRONICS

SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. A logic gate is electronic circuit which
   a) Alternate between 0 and I values
   b) Works on binary algebra
   c) Allows electron flow only in one direction
   d) Makes logic decision
2. The output of a 2-input OR gate is zero only when
   a) Either input is 0
   b) Both inputs are 1
   c) Both inputs are 0
   d) Either input is 1
3. The number 1110_2 is equivalent to decimal number
   a) One thousand
   b) Four
   c) Sixteen
   d) Fourteen
4. Digital system operates on which number system
   a) Binary
   b) Decimal
   c) Octal
   d) Hexadecimal
5. The frequency of oscillation of an astable multivibrator depends mainly on
   a) Resistor-capacitor values of the circuit
   b) Value of collector resistor
   c) Value of transistor gain
   d) Width of input pulse
6. An example of a non-sinusoidal oscillator is
   a) Hartley oscillator
   b) Bistable multivibrator
   c) Colpitts oscillator
   d) Tuned oscillator
7. An oscillator differs from an amplifier because it
   a) Requires no input signal
   b) Has more gain
   c) Requires no dc supply
   d) Always has the input and output
8. An astable multivibrator has
   a) One stable state
   b) Two stable states
   c) No stable state
   d) Three stable states

9. An XOR gate has an output only when
   a) Both input is at logic 1
   b) Any input is at logic 1 but not both
   c) Both inputs are at logic o
   d) Any input is at logic o but not both
10. A multivibrator has how many stable states
   a) One
   b) Two
   c) None
   d) All of the above
11. When an SCR is forward biased, what is needed to cause it conduct
   a) A drain signal
   b) A source current
   c) A source-drain signal
   d) A gate signal
12. In a class A amplifier the output current flows for
   a) A part of the cycle of the input signal
   b) The full cycle of the input signal
   c) Half the cycle of the input signal
   d) ¾ of the circle of the input signal
13. _____ is the device which is fabricated as two SRC's in parallel with common gate terminal
   a) Field effect transistor
   b) Triac
   c) Diac
   d) Unijuction transistor
14. A thermocouple is a _____type of transducer
   a) Variable resistance
   b) Variable inductance
   c) Voltage generating
   d) Voltage divider
15. An SCR is turned off by _____
   a) Interrupting its anode current
   b) Reversing polarity of its anode-cathode voltage
   c) Reduce anode current
   d) All of the above
16. The output of a class A amplifier is
   a) Is not distortion free
   b) Consists of positive pulses only
   c) Is like output of a full wave rectifier
   d) Comprises of short duration pulses

17. _____ is two transistors with a control element
   a) Diac
   b) Thyristor
   c) Triac
   d) Diode
18. An operational amplifier can be connected as the following except
   a) Inverting amplifier
   b) Non-inverting amplifier
   c) Oscillator
   d) Summing amplifier
19. The following amplifier is used for amplifying low frequency signals
   a) Audio amplifier
   b) T.V amplifier
   c) D.C amplifier
   d) Power amplifier
20. When a transistor is driven into saturation ideally the output is
   a) 2Vcc
   b) 3Vcc
   c) ½ Vcc
   d) Vcc
21. Which of the following statements is true about the field effect transistor?
   a) It has a high output impedance
   b) It does not offer any resistance
   c) It has a low input impedance
   d) It has a high input impedance
22. Comparing the size of the BJT and the FET
   a) BJT is larger than the FET
   b) They are the same size
   c) BJT is smaller than the FET
   d) Depends on application
23. Enhancement only MOSFET refers to a set which
   a) Operates only in enhancement mode
   b) Operates only with positive voltage
   c) Operates only with negative voltage
   d) Operates both negative and positive mode
24. Improper biasing of a transistor circuit leads to
   a) Excessive heat production
   b) Distortion in the output signal
   c) Faulty location of load line
   d) Heavy loading of the transistor

25. Transistor amplifiers use _____ feedback
    a) Positive feedback
    b) Negative feedback
    c) Both positive and negative feedback
    d) No feedback
26. The following statements are true about BJTS. Which one is false
    a) Base is thinly doped
    b) Collector is wider and thickly doped
    c) Base is sandwiched between collector and emitter
    d) Emitter current is equal to collector current
27. Majority carriers in an PNP transistor are
    a) Holes
    b) Electrons
    c) Neutrons
    d) Both electrons and holes
28. Transistor as a switch operates at
    a) Cut off region only
    b) Saturation region only
    c) Between cut off and saturation region
    d) Active region only
29. The channel of a JFET is between
    a) Gate and drain
    b) Drain and source
    c) Gate and source
    d) Input and outputs
30. A single stage transistor amplifier contains _____ and associated circuitry.
    a) Two transistors
    b) One transistor
    c) Three transistors
    d) None of the above
31. Which one of the following is the correct definition of power rating of a resistor?
    a) It's the maximum power that a resistor can dissipate as heat without being damaged
    b) It indicates the minimum power of a resistor
    c) It is the change in resistance of a resistor
    d) It is the tolerance of a resistor
32. The following are factors considered while selecting a capacitor, which one is **not**?
    a) Tolerance
    b) Stability
    c) Leakage current
    d) Capacitive reactance

33. The output of unfiltered half-wave is
    a) Steady dc voltage
    b) Pulsating dc voltage
    c) Smooth dc voltage
    d) None of the above
34. What is the function of a zener diode in power supply circuit?
    a) An amplifier
    b) Rectifier
    c) Voltage regulator
    d) Resistor
35. What is extrinsic conduction?
    a) Application of heat to a pure semi-conductor
    b) Introduction of an external impurity to a pure semi-conductor
    c) It is a forward biased P-N junction
    d) It is reversely biased P-N junction
36. An ideal PN junction diode is one which behaves as a perfect_____ when forward biased.
    a) Conductor
    b) Insulator
    c) Resistance material
    d) None of the above
37. In the breakdown region, zener diode behaves like _____ source
    a) Constant resistance
    b) Constant power
    c) Constant voltage
    d) Constant current
38. In reverse biased PN junction _____ carriers cause leakage current?
    a) Majority carriers
    b) Minority carriers
    c) Both majority and minority carriers
    d) Either the majority or minority carriers
39. The transducer that convert the input signal into the output signal, which is a discrete function of time is known as _____
    a) Active
    b) Analogue
    c) Digital
    d) Pulse
40. When a trivalent impurity is added to a semiconductor trivalent becomes _____of electrons
    a) Donor
    b) Acceptor
    c) P- type
    d) N-type

SECTION 2 SHORT ANSWER QUESTIONS (40 MARKS)

1. Name the five gates commonly used and state their differences. (5marks)
2. Draw a labelled block diagram of an inductor-capacitor Oscillator(LC). (5marks)
3. a) State any **three** characteristics of an ideal OP- amp. (3marks)
   b) State any **two** applications of transducers. (2marks)
4. Name any **three** classes of amplifier and state their differences. (5marks)
5. a) State the **two** types of MOSFETS and one application of FETS. (3marks)
   b) state any **two** advantages of FETS compared to BJTS. (2marks)
6. Draw the output characteristics curve of a BJT indicating:
   - Saturation region
   - Active region
   - Cut-off region (5 marks)
7. a) Name and sketch any two types of transistor circuit configuration. (4 marks)
   b) State the most commonly used configuration. (1 mark)
8. a) State the value of resistors whose colors are:
   i. Green, blue, gold and silver
   ii. Gray, red, silver and gold (2marks)

   b) State any three applications of capacitors. (3marks)

SECTION 3 LONG ANSWER QUESTIONS (20 MARKS)

1. a) Draw a labelled diagram of a centre-tapped full-wave rectifier and explain how the circuit works. (10marks)

   b) name and explain the functions of the terminals of a JFET. (6marks)
   c) Sketch a resistance temperature graph of a thermistor and explain its shape. (4 marks)